



Parallel & Distributed Computing

Lecture NO: 01

Introduction

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Course Pre-requisites

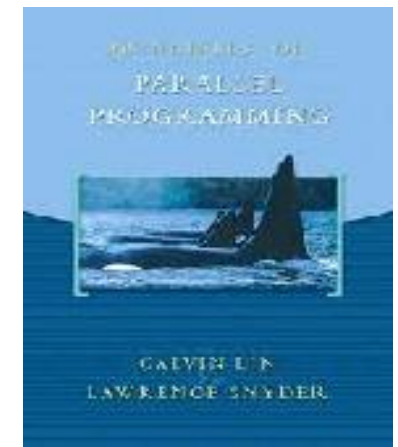
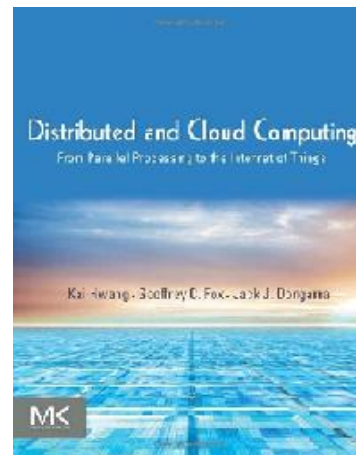
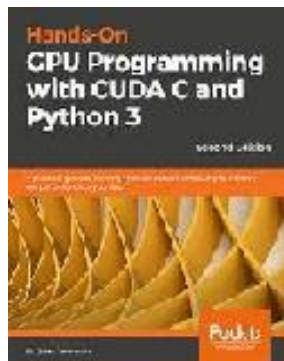
- Programming Experience (preferably Python/C++/Java)
- Understanding of Computer Organization and Architecture
- Understanding of Operating System

Requirements & Grading

- Roughly
 - 50 % Final Exam
 - 25% Internal Evaluation
 - Quiz 8 Marks
 - Assignments 8 Marks
 - Project 9 Marks
 - 25% Mid term exam

Books

- Some good books are:
 - Distributed Systems Third edition
 - PRINCIPLES OF PARALLEL PROGRAMMING
 - Designing and Building Parallel Programs
 - Distributed and Cloud Computing



Course Project

- At the end of the semester students need to submit a semester project like
 - Distributed computing & smart city services
 - Large scale convolutional neural networks
 - Distributed computing with delay tolerant network

Course Overview

- This course covers following main concepts
 - Concepts of parallel and distributed computing
 - Analysis and profiling of applications
 - Shared memory concepts
 - Distributed memory concepts
 - Parallel and distributed programming (OpenMP, MPI)
 - GPU based computing and programming (CUDA)
 - Virtualization
 - Cloud Computing, MapReduce
 - Grid Computing
 - Peer-to-Peer Computing
 - Future trends in computing

Recommended Material

- Distributed Systems, Maarten van Steen & Andrew S. Tanenbaum, 3rd Edition (2020), Pearson.
- Parallel Programming: Concepts and Practice, Bertil Schmidt, Jorge Gonzalez-Dominguez, Christian Hundt, Moritz Schlarb, 1st Edition (2018), Elsevier.
- Parallel and High-Performance Computing, Robert Robey and Yuliana Zamora, 1st Edition (2021).
- Distributed and Cloud Computing: From Parallel Processing to the Internet of Things, Kai Hwang, Jack Dongarra, Geoffrey Fox, 1st Edition (2012), Elsevier.
- Multicore and GPU Programming: An Integrated Approach, Gerassimos Barlas, 2nd Edition (2015), Elsevier.
- Parallel programming: For multicore and cluster systems. Rauber, Thomas, and Gudula Rünger. Springer Science & Business Media, 2013.

Recent Jobs

Software Engineer in Machine Learning Systems Graduate (AML - Machine Learning...

ByteDance

San Jose, CA 95110 (Downtown area)

\$112,200 - \$147,000 a year

- Master **distributed, parallel computing** principles; know the recent **advances** in **computing**, storage, networking, and hardware technologies;

Posted 30 days ago · [More...](#)

[View all 8 available locations](#)

Data Engineer

CivicPlus

Remote

 Estimated \$85.8K - \$109K a year ?

Full-time

On call

- Understanding of **distributed computing** architectures and **parallel** processing techniques.
- Your expertise in data engineering will be instrumental in ensuring the...

Posted 17 days ago · [11 25+ applications](#) · [More...](#)

Jobs

Associate Distributed Systems Engineer

Arcadia.io

Remote

 Estimated \$87.5K - \$111K a year ?

Full-time

- Engineers need to ensure that tasks are efficiently **distributed** across the system's resources to make the most of available **computing** power.

Posted 30+ days ago •  25+ applications · More...

Virtualization Engineer

Absolute Business Solutions Corp

Bethesda, MD 20814

 Estimated \$121K - \$153K a year ?

- Using common **distributed** and **parallel** processing frameworks: MapReduce, Bulk Synchronous **Parallel**, MapReduce, Bulk Synchronous **Parallel**, Online Analytic Graph...

Posted 30+ days ago •  25+ applications · More...

Research In Parallel & Distributed Computing

Top 4 Research Topics

- ✓ Geo distributed fog computing
- ✓ Multi clustering approach
- ✓ Service attacks in SDN
- ✓ Distributed computing in DTN
- ✓ Large scale in CNN



Worldwide
Service



Dedicated
Team



Domain
Expert



Prime
Quality

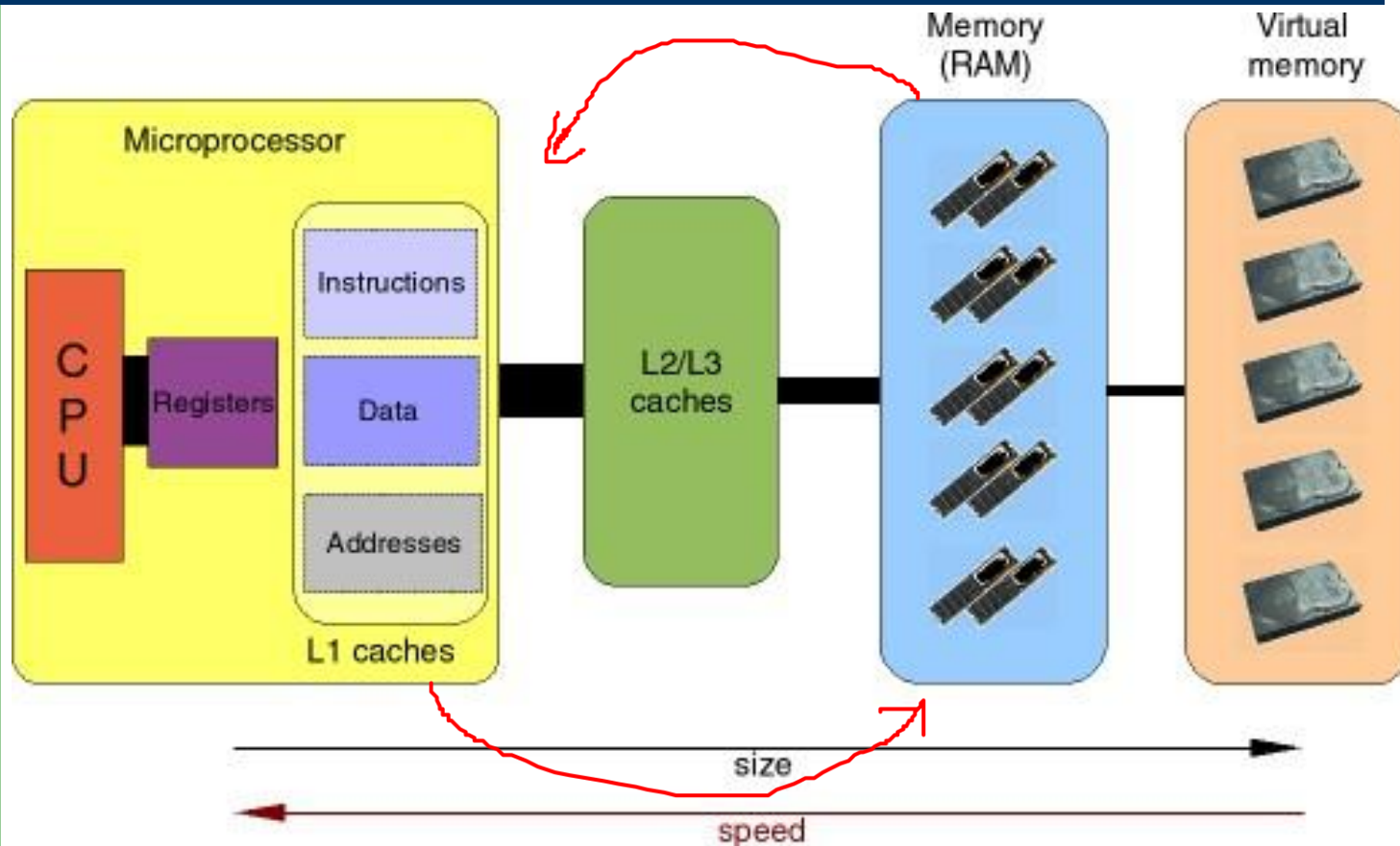


Research Topics in Parallel and
Distributed Computing

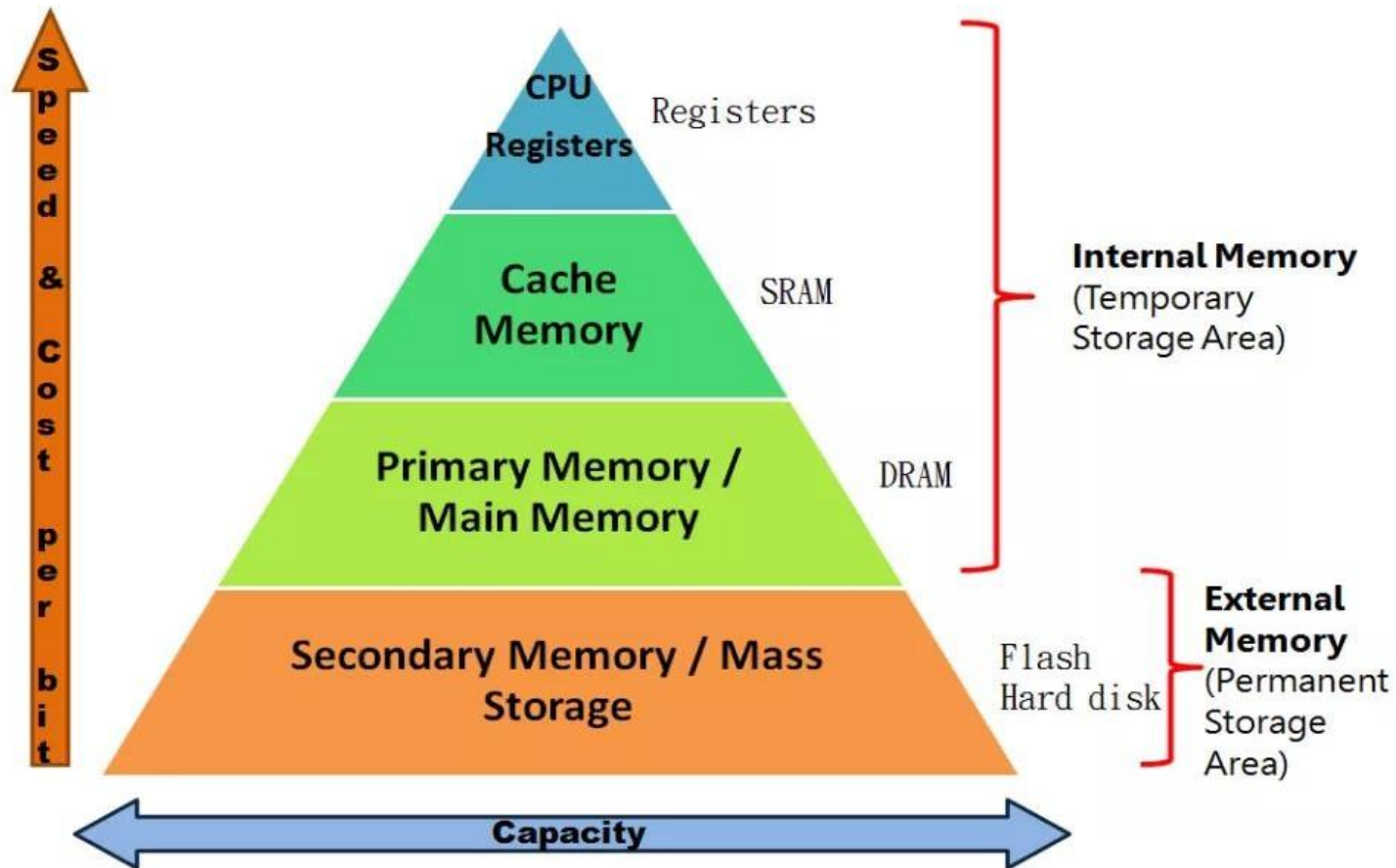


www.networksimulationtools.com

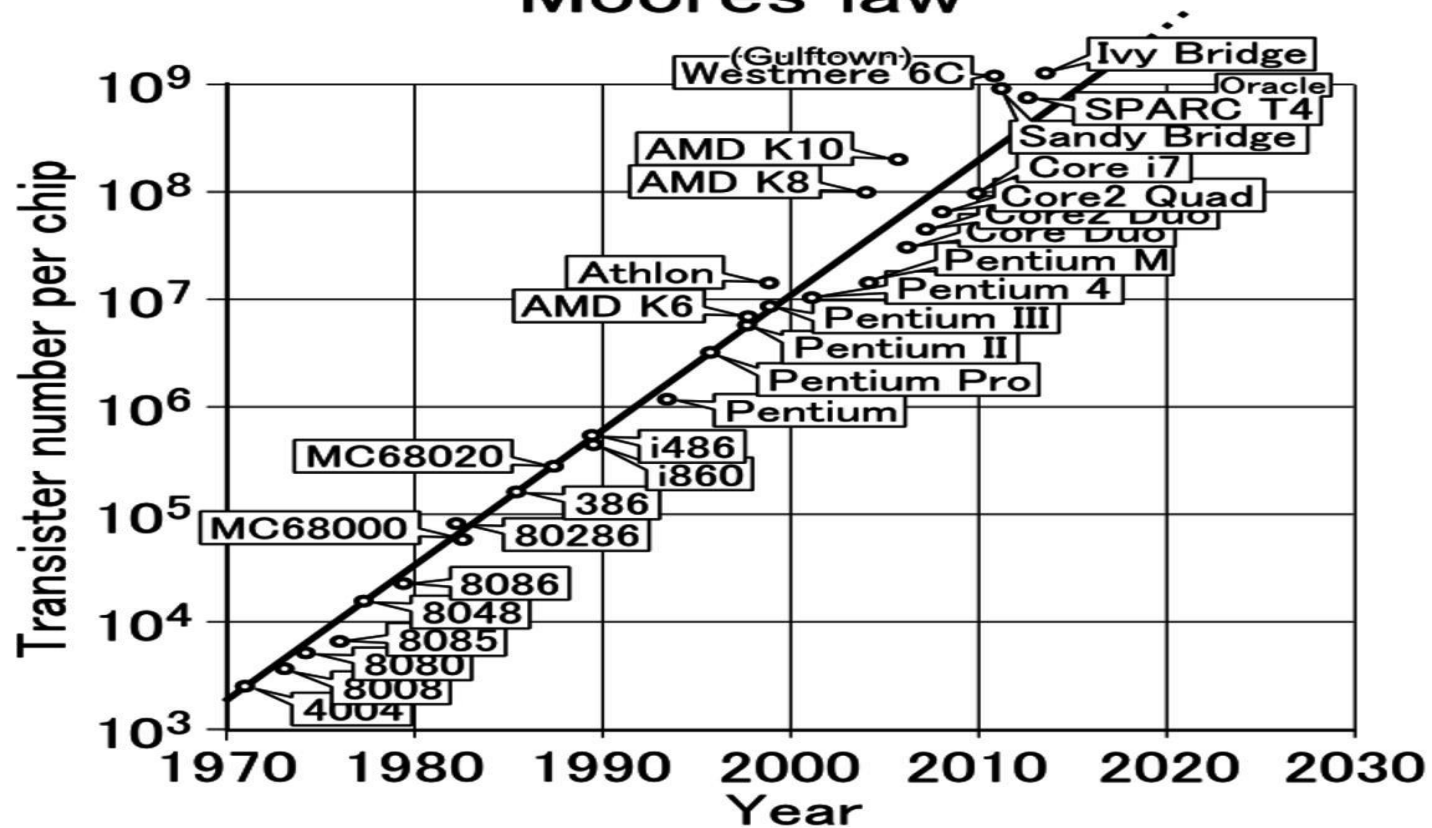
Single Processor Architecture



Memory Hierarchy



Moore's law



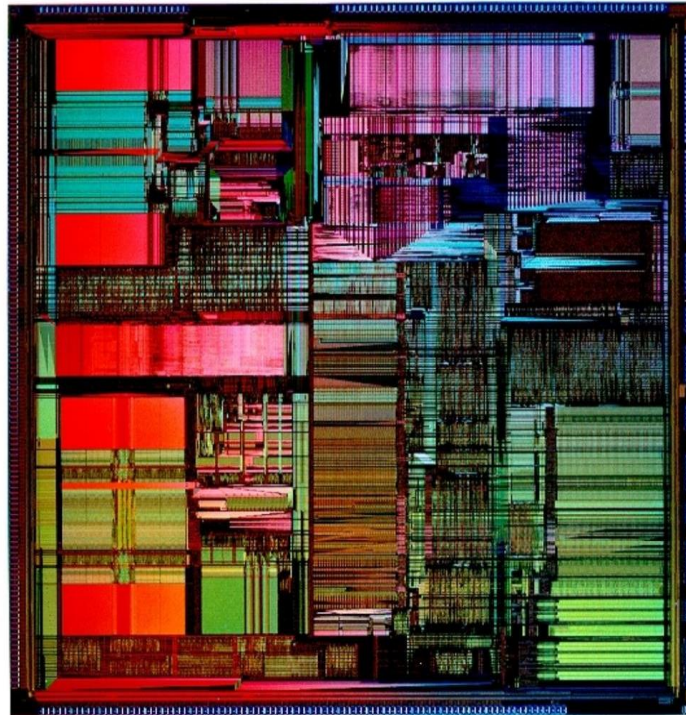
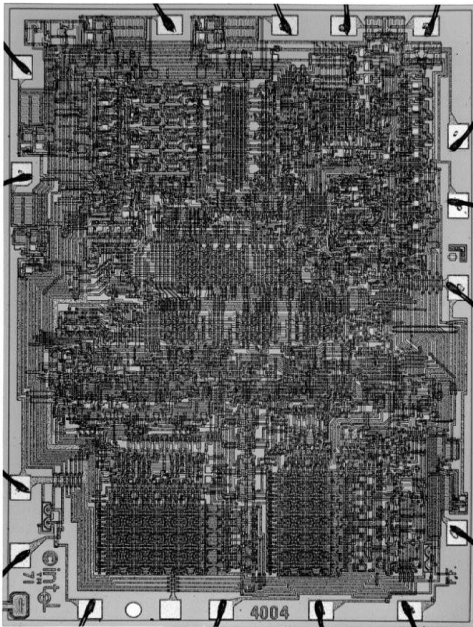
5 years of Technology Advance

~~INTEL 4004 (1971)~~

4-bit data

2300 transistors, 10 microns

0,06 MOPS, 108 kHz



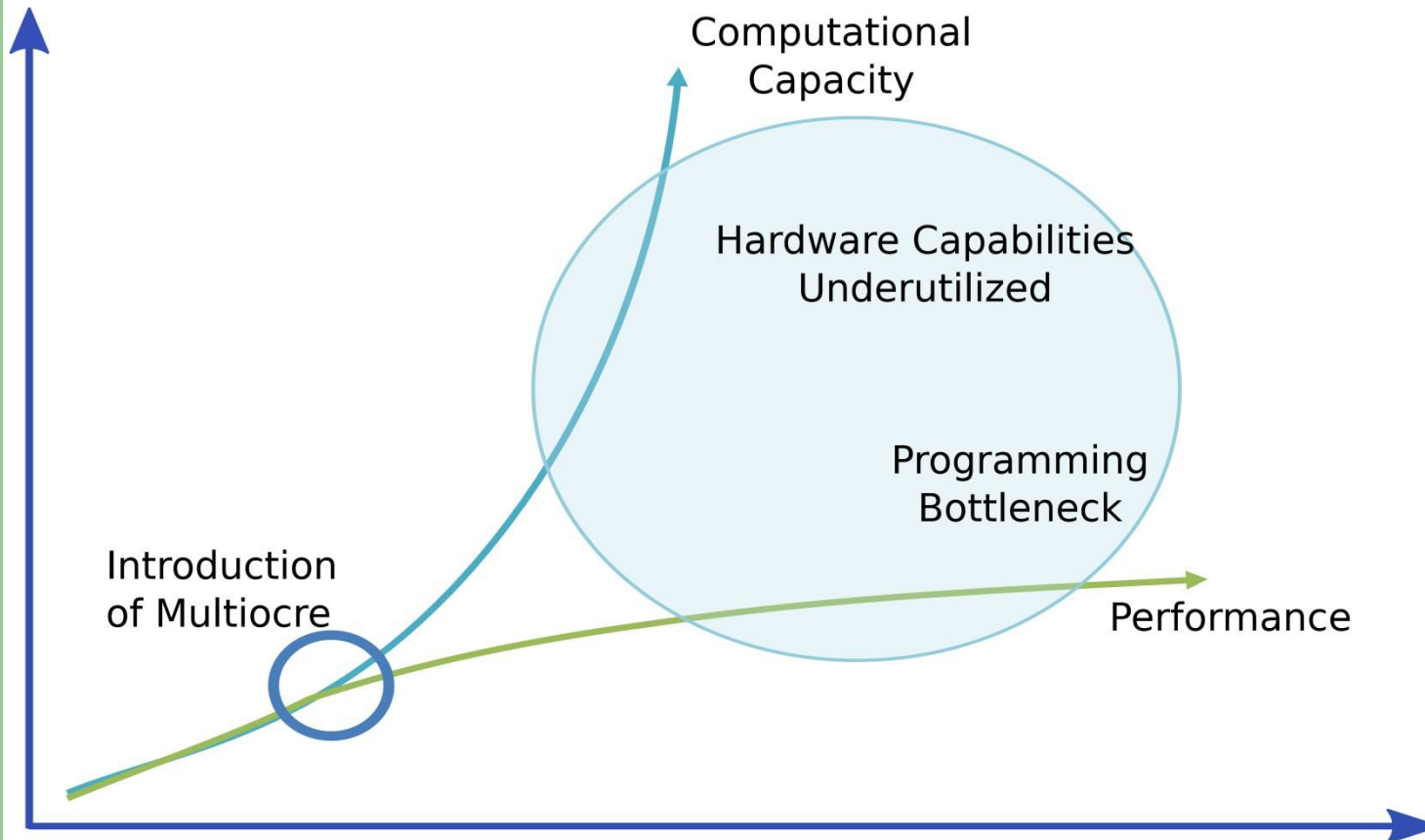
INTEL Pentium II (1996)

32-bit data

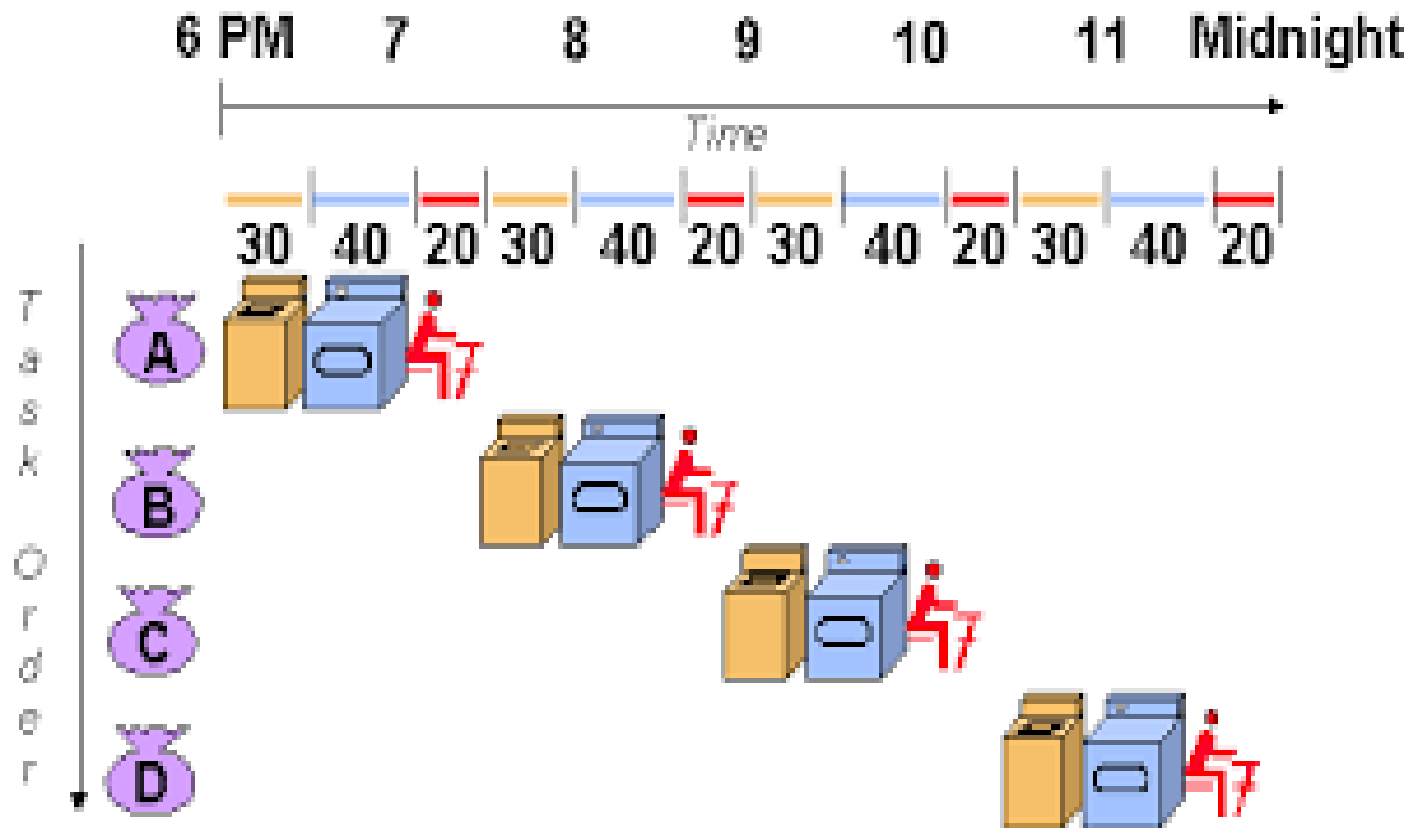
5.5M transistors, 0.35 μ , 2 cm²

200 MHz, 200 MOPS, 3.3V, 35W

Productivity Gap

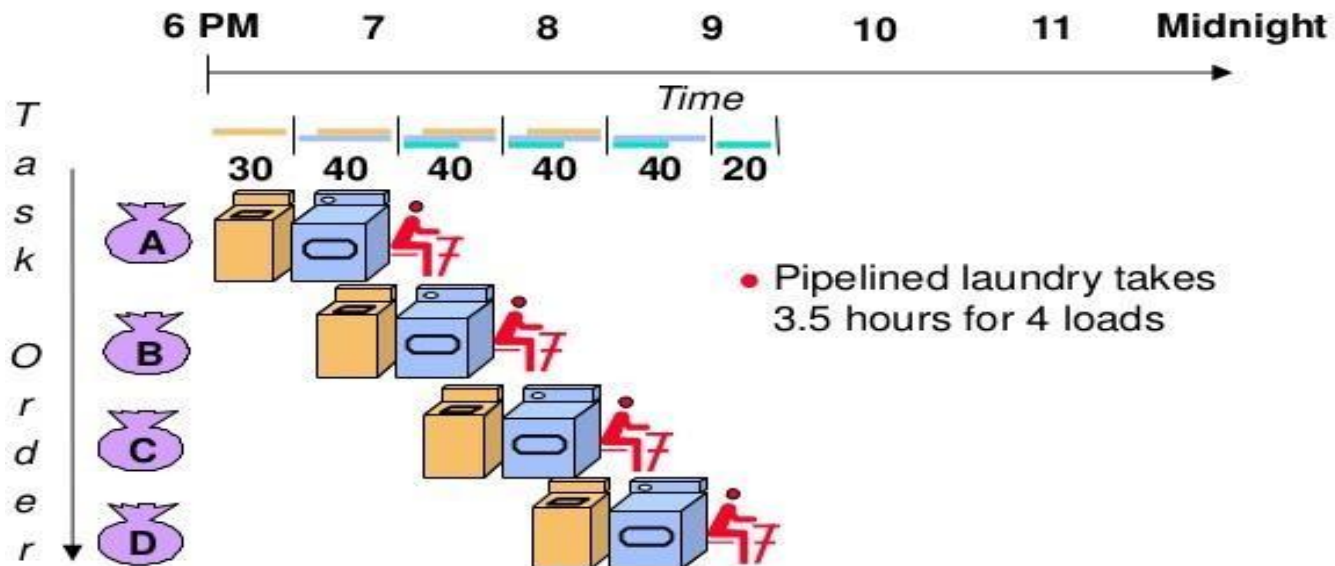


Pipelining



Pipelining

Traditional Pipeline Concept

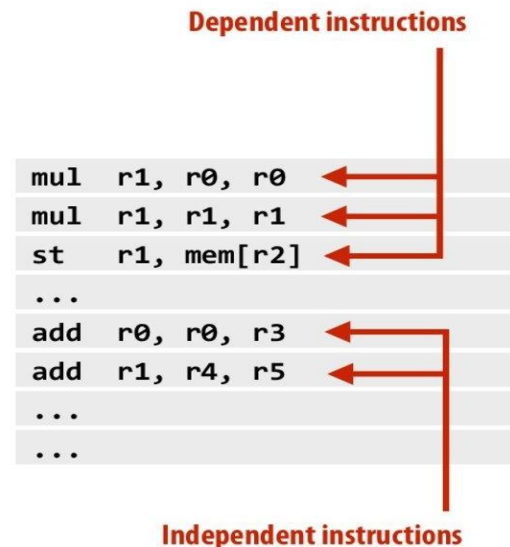


Until 10 years ago: two significant reasons for processor performance improvement

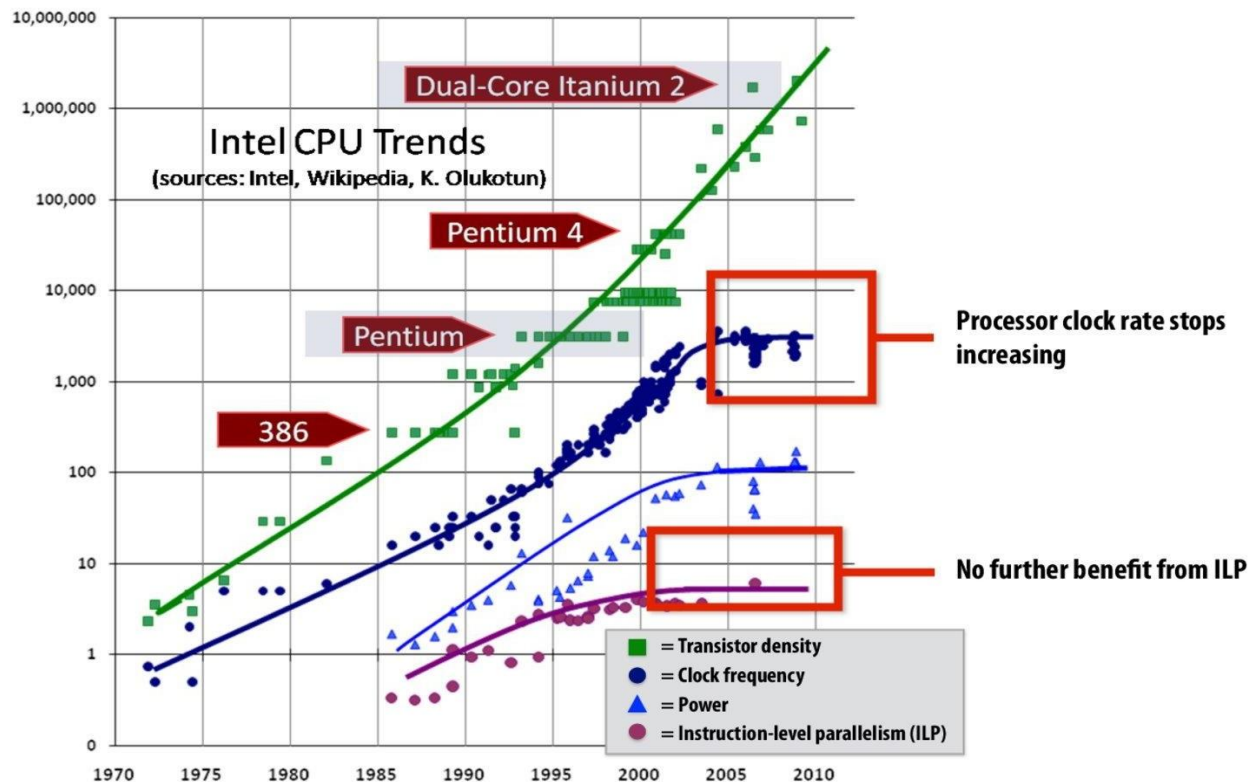
- 1. Increasing clock frequency**
- 2. Exploiting instruction-level parallelism (superscalar execution)**

Instruction level parallelism (ILP)

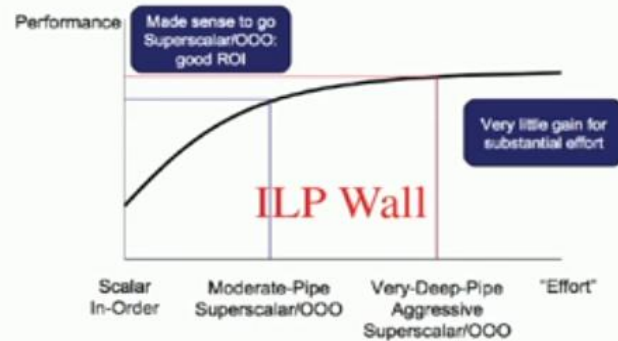
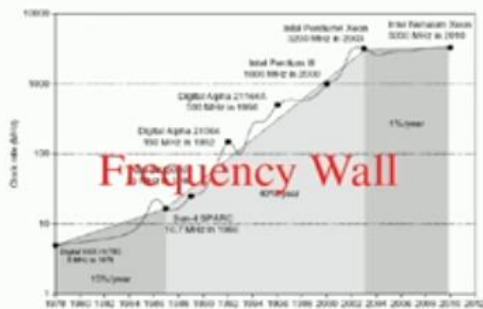
- Processors did in fact leverage parallel execution to make programs run faster, it was just invisible to the programmer
- Instruction level parallelism (ILP)
 - Idea: Instructions must appear to be executed in program order. BUT independent instructions can be executed simultaneously by a processor without impacting program correctness
 - Superscalar execution: processor dynamically finds independent instructions in an instruction sequence and executes them in parallel



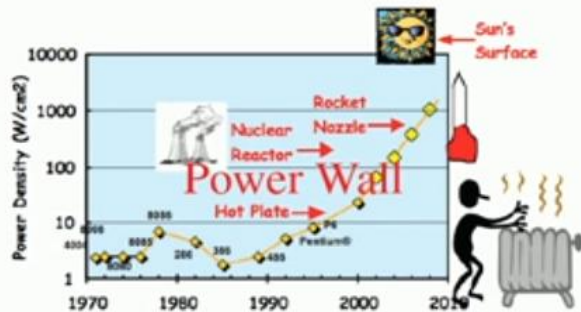
ILP tapped out + end of frequency scaling



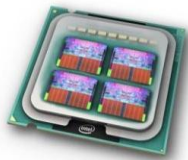
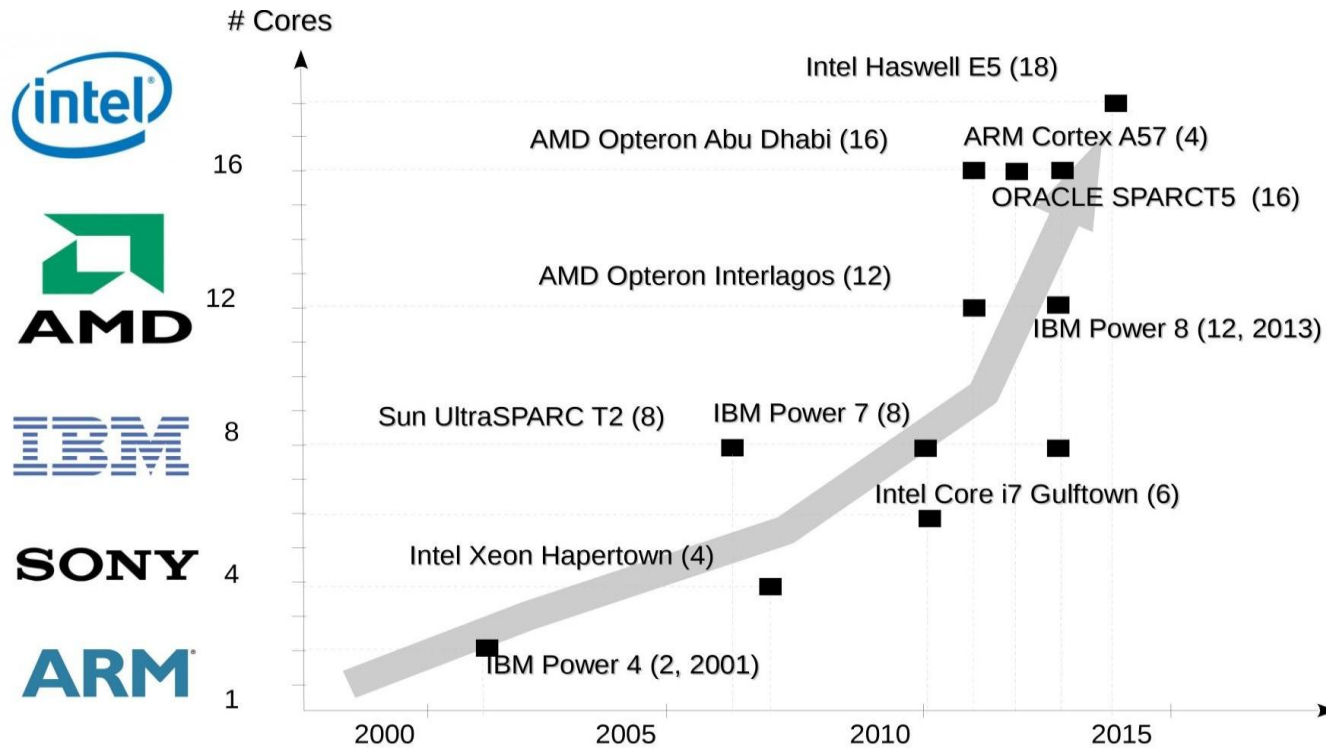
Roadblocks for Single-Core Performance



Source: G. Loh

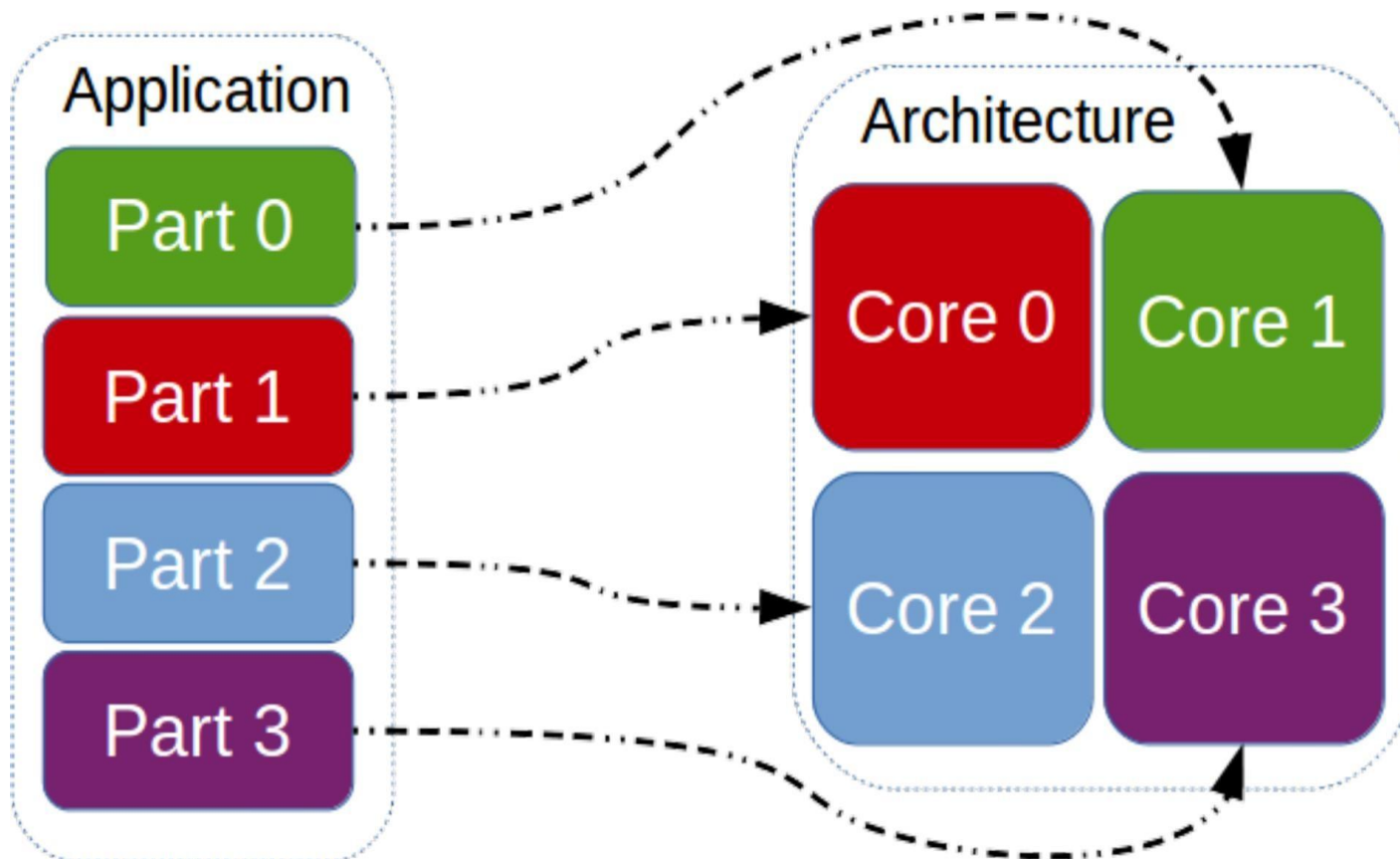


Multicore Trend



Multicore	Multiprocessor
Single or multiple integrated circuit die/s	Single or multiple systems
Cheaper (single CPU that does not require multiple CPU support system)	Expensive (Multiple separate CPU's that require a system that supports multiprocessors)
Will have less traffic (cores integrated into a single chip and will require less time)	Will have more traffic (distances between the two will require a longer time)
Does not need to be configured	Needs a little complex configuration
Faster running a single program	Faster running multiple programs

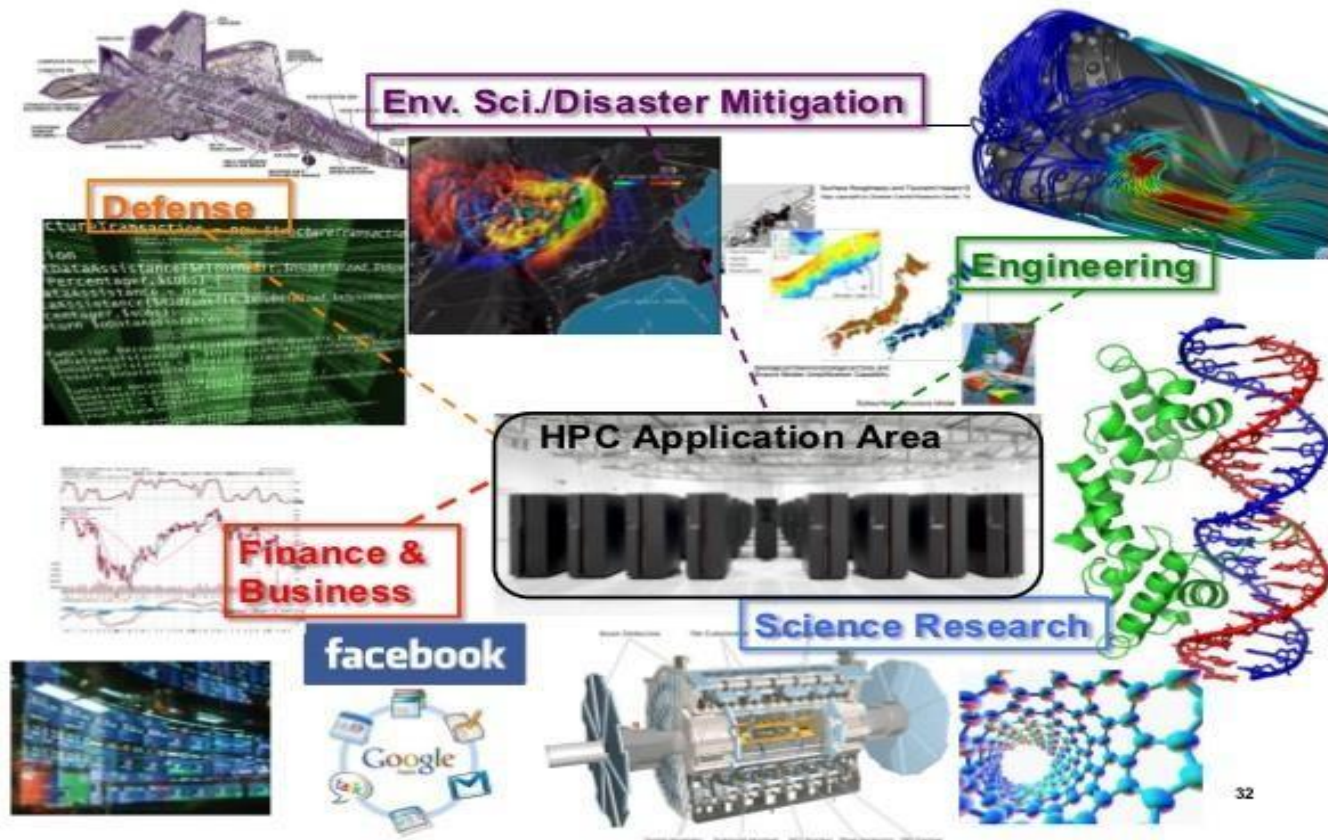
Application Partitioning



High-Performance Computing (HPC)

- HPC is the use of parallel processing for running advanced application programs efficiently, reliably and quickly.
- It applies especially to systems that function above a tera FLOPs (floating-point operations per second) processing speed.
- The term HPC is occasionally used as a synonym for supercomputing, although technically a supercomputer is a system that performs at or near the currently highest operational rate for computers.

High Performance Computing



GPU-accelerated Computing

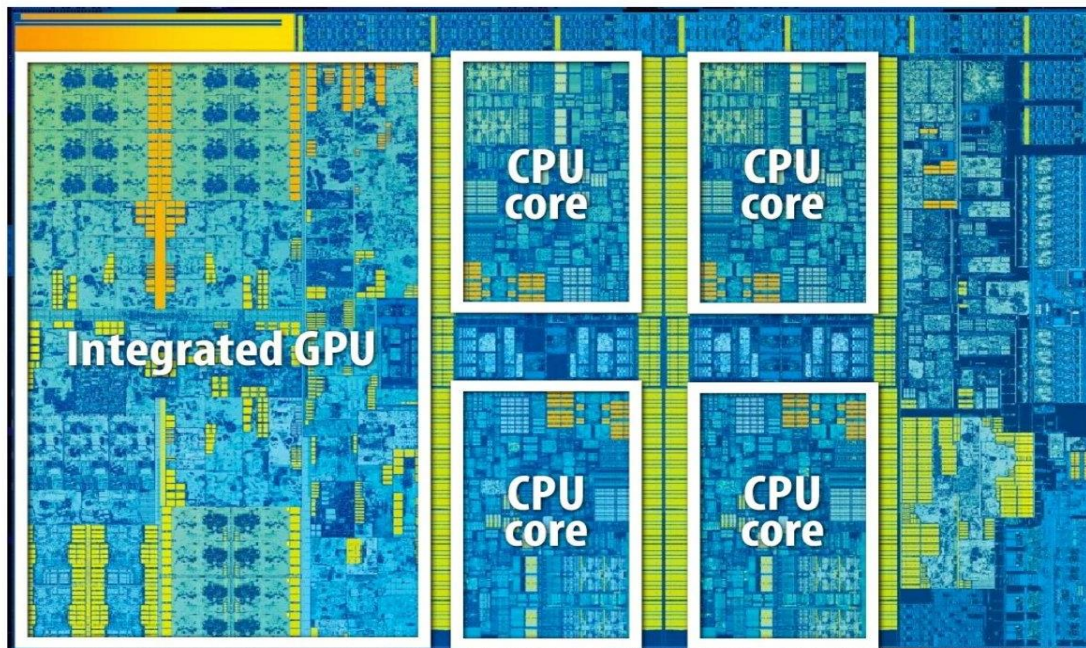
- GPU-accelerated computing is the use of a graphics processing unit (GPU) together with a CPU to accelerate deep learning, analytics, and engineering applications.
- Pioneered in 2007 by NVIDIA, GPU accelerators now power energy-efficient data centers in government labs, universities, enterprises, and small-and-medium businesses around the world.
- They play a huge role in accelerating applications in platforms ranging from artificial intelligence to cars, drones, and robots.

What is GPU?

- It is a processor optimized for 2D/3D graphics, video, visual computing, and display.
- It is highly parallel, highly multithreaded multiprocessor optimized for visual computing.
- It provide real-time visual interaction with computed objects via graphics images, and video.
- It serves as both a programmable graphics processor and a scalable parallel computing platform.
- Heterogeneous Systems: combine a GPU with a CPU

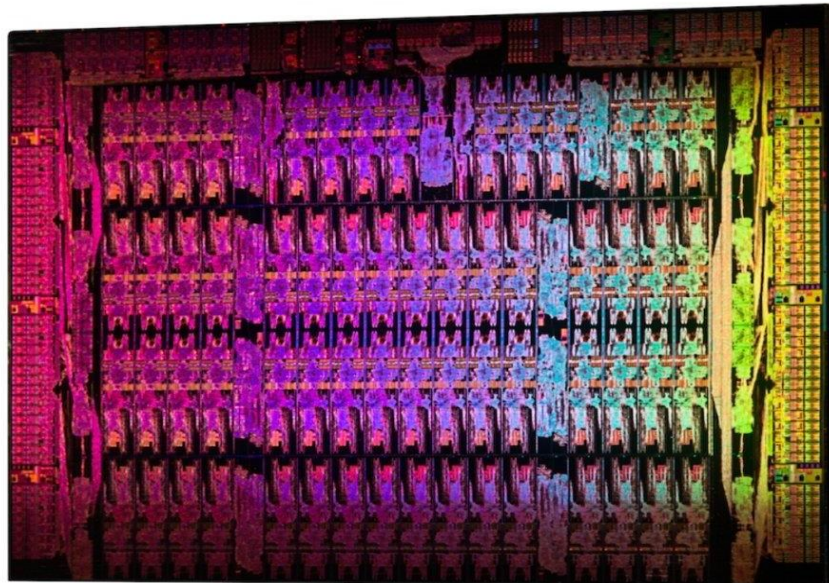
Intel Skylake (2015) (aka “6th generation Core i7”)

Quad-core CPU + multi-core GPU integrated on one chip



Intel Xeon Phi 7120A “coprocessor”

- 61 “simple” x86 cores (1.3 Ghz, derived from Pentium)
- Targeted as an accelerator for supercomputing applications





NVIDIA Maxwell GTX 980 GPU (2014)

Sixteen major processing blocks
(but much, much more parallelism available... details coming next class)

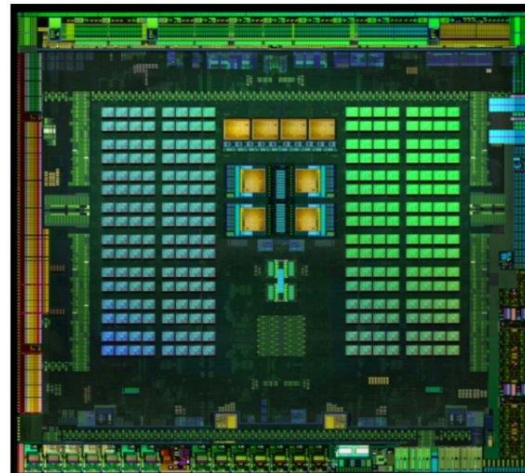


Mobile parallel processing

Power constraints heavily influence design of mobile systems



Apple A9: (in iPhone 6s)
Dual-core CPU + GPU + image processor
and more on one chip



NVIDIA Tegra K1:
Quad-core ARM A57 CPU + 4 ARM A53 CPUs +
NVIDIA GPU + image processor...

SGI Altix Supercomputer 2300 processors

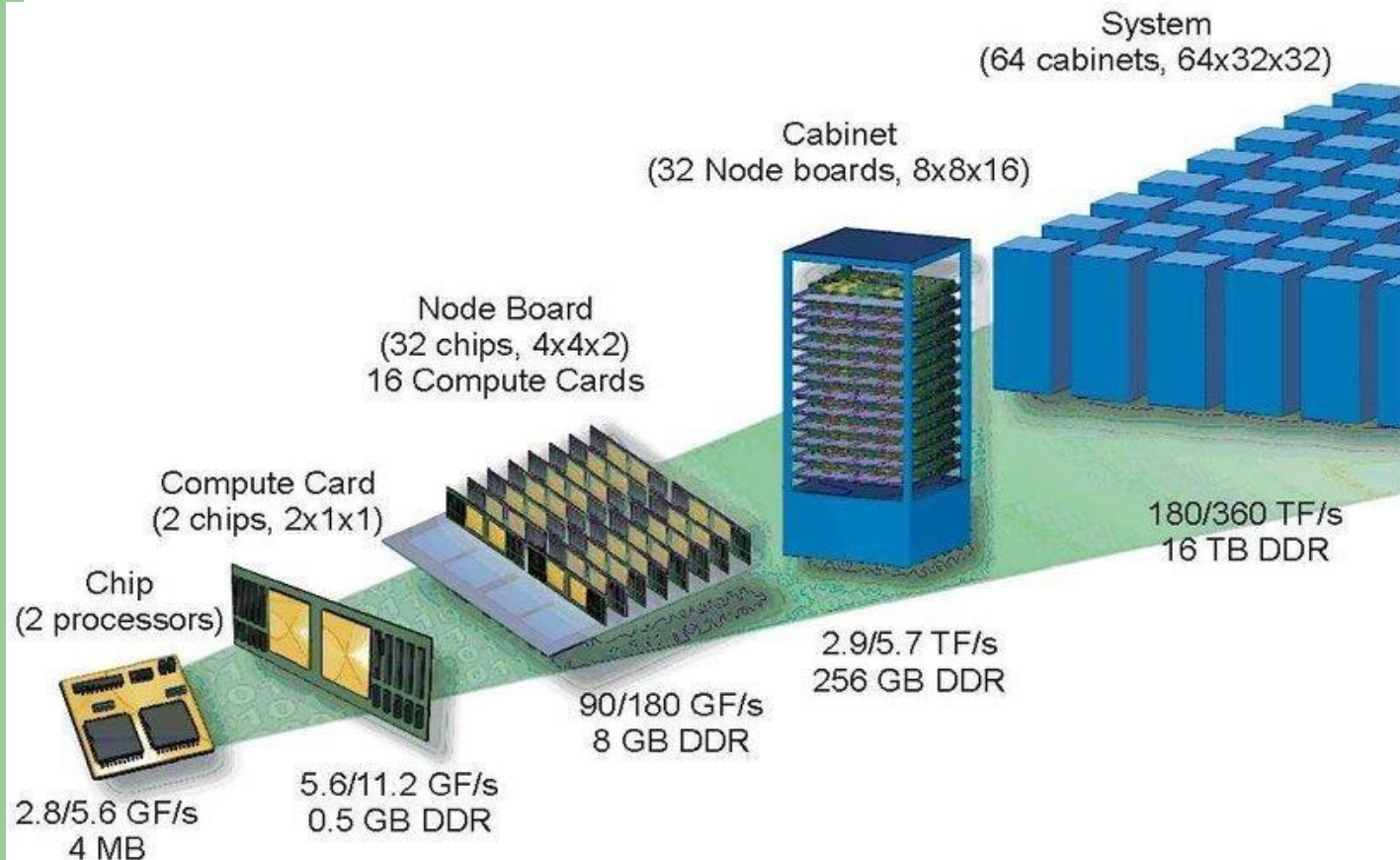


Supercomputing

- Today: clusters of multi-core CPUs + GPUs
- Oak Ridge National Laboratory: Titan (#2 supercomputer in world)
 - 18,688 x 16 core AMD CPUs + 18,688 NVIDIA K20X GPUs

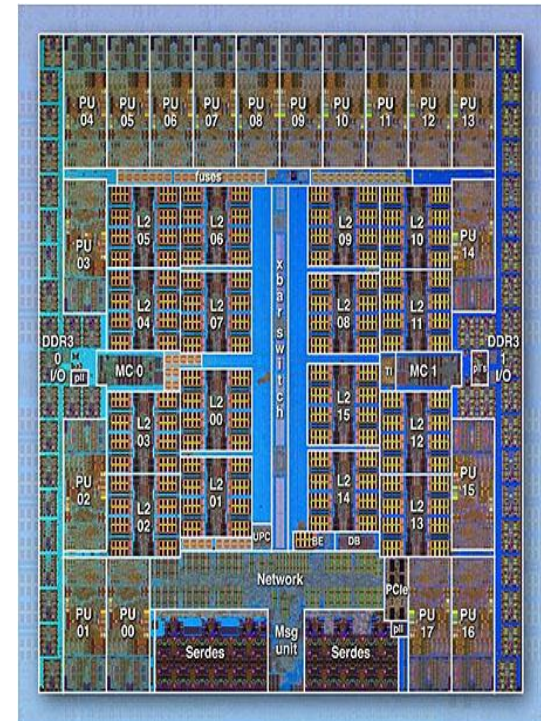


HPC System composition



Parallel Computers

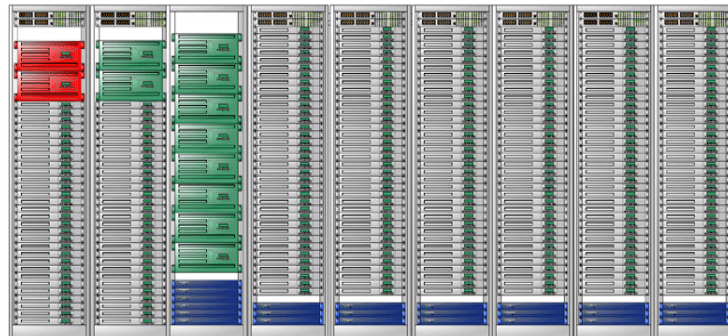
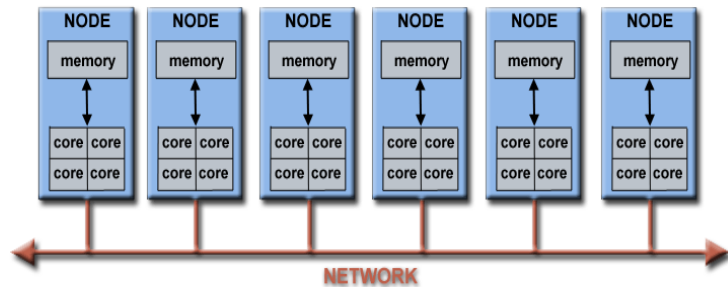
- Virtually all stand-alone computers today are parallel from hardware perspective:
 - Multiple functional units (L1 cache, L2 cache, branch, pre-fetch, decode, floating-point, graphics processing (GPU), integer, etc.)
 - Multiple execution units/cores
 - Multiple hardware threads



IBM BG/Q Compute Chip with 18 cores (PU) and 16 L2 Cache units (L2)

Parallel Computers

- Networks connect multiple stand-alone computers (nodes) to make larger parallel computer clusters.
- Parallel computer cluster
 - Each compute node is a multi-processor parallel computer in itself
 - Multiple compute nodes are networked together with an Infiniband network
 - Special purpose nodes, also multi-processor, are used for other purposes

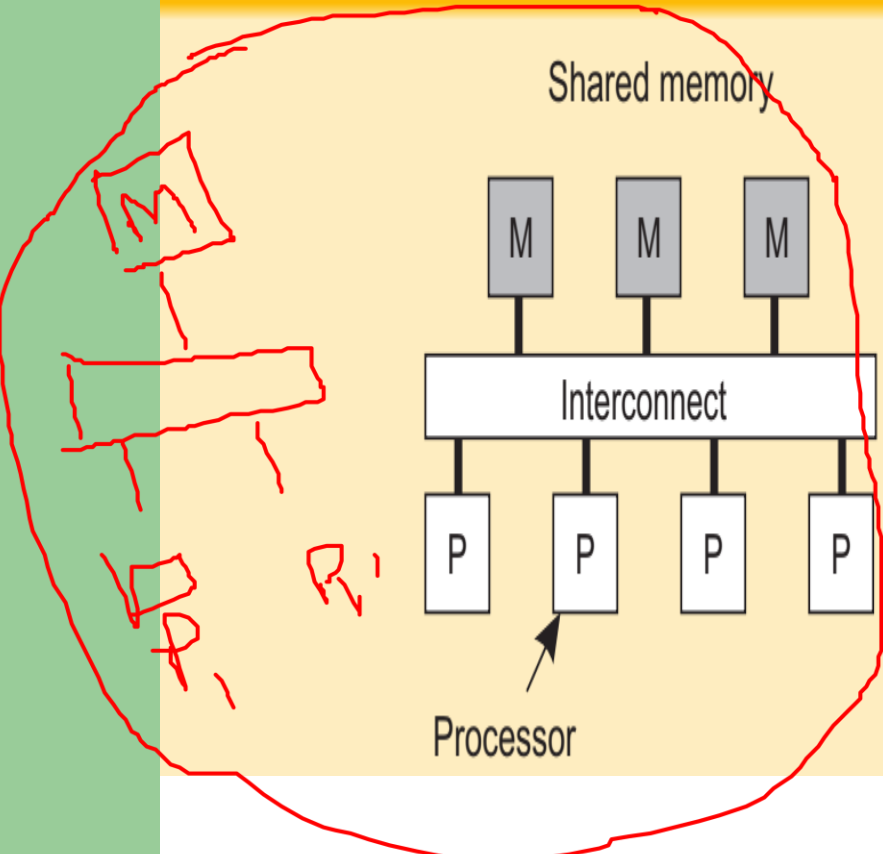


Types of Parallel and Distributed Computing

- Parallel Computing
 - Shared Memory
 - Distributed Memory
- Distributed Computing
 - Cluster Computing
 - Grid Computing
 - Cloud Computing
 - Distributed Pervasive Systems

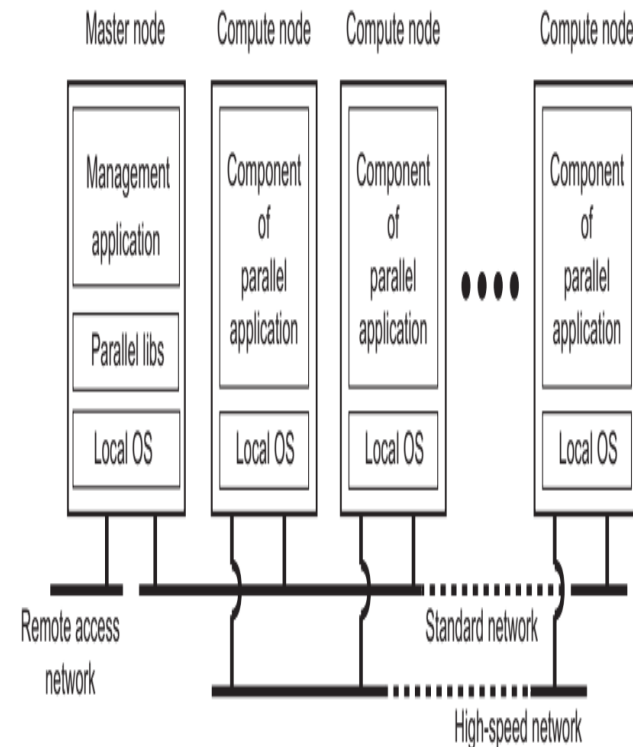
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Multiprocessor and multicore versus multicomputer



Distributed (Cluster) Computing

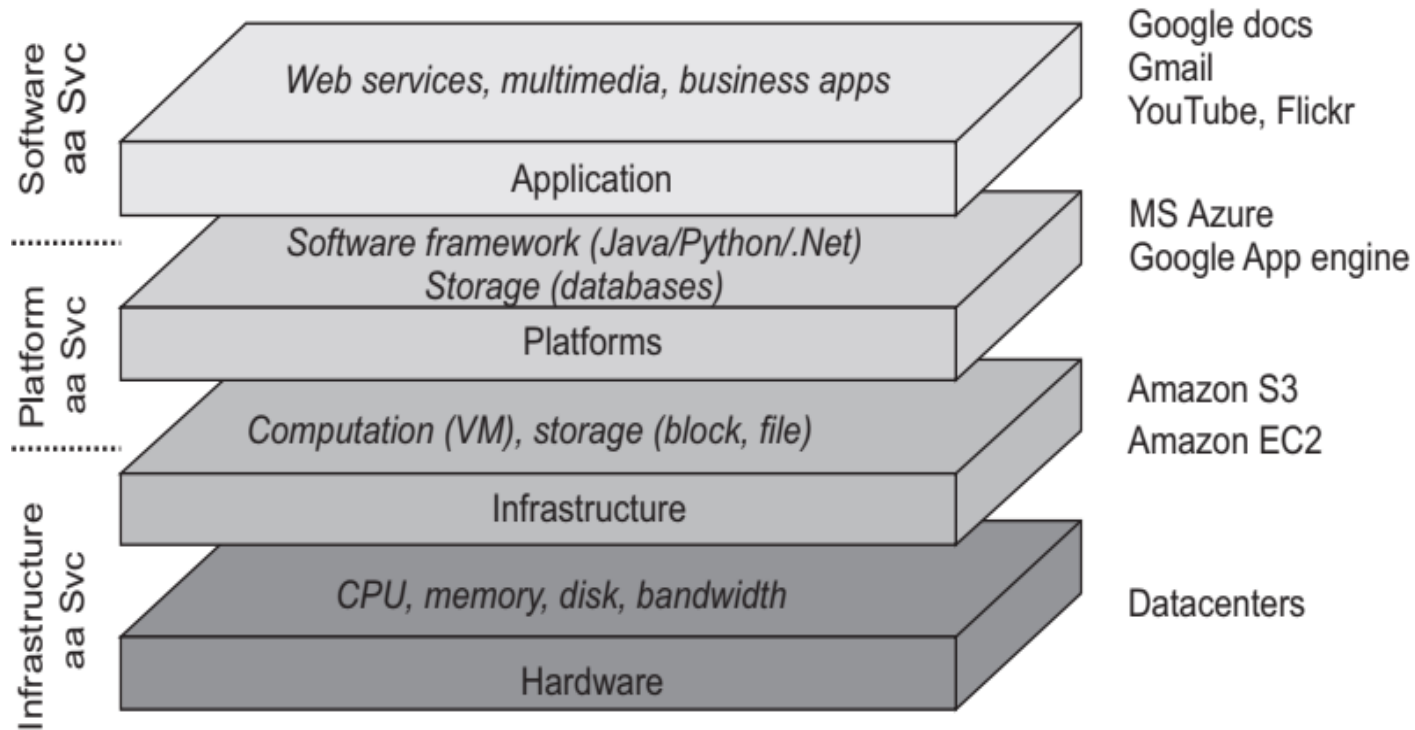
- Essentially a group of high-end systems connected through a LAN
- Homogeneous: same OS, near-identical hardware
- Single managing node



Distributed (Grid) Computing

- Lots of nodes from everywhere
 - Heterogeneous
 - Dispersed across several organizations
 - Can easily span a wide-area network
- To allow for collaborations, **grids** generally use virtual organizations.
- In essence, this is a grouping of users (or their IDs) that will allow for authorization on resource allocation.

Distributed (Cloud) Computing



Distributed (Pervasive) Computing

- Emerging next-generation of distributed systems in which nodes are small, mobile, and often embedded in a larger system, characterized by the fact that the system naturally blends into the user's environment.
- Three subtypes
 - **Ubiquitous computing systems:** pervasive and continuously present, i.e., there is a continuous interaction between system and user.
 - **Mobile computing systems:** pervasive, but emphasis is on the fact that devices are inherently mobile.
 - **Sensor (and actuator) networks:** pervasive, with emphasis on the actual (collaborative) sensing and actuation of the environment.

Why Use Parallel Computing?



The Real World is Massively Parallel

- In the natural world, many complex, interrelated events are happening at the same time, yet within a temporal sequence.
- Compared to serial computing, parallel computing is much better suited for modeling, simulating and understanding complex, real world phenomena.
- For example, imagine modeling these serially =>



Rush Hour Traffic



Plate Tectonics



Weather



Auto Assembly



Jet Construction



Drive-thru Lunch



Galaxy Formation



Planetary Movements



Climate Change

SAVE TIME AND/OR MONEY (Main Reasons)

- In theory, throwing more resources at a task will shorten its time to completion, with potential cost savings.
- Parallel computers can be built from cheap, commodity components.



SOLVE LARGER / MORE COMPLEX PROBLEMS (Main Reasons)

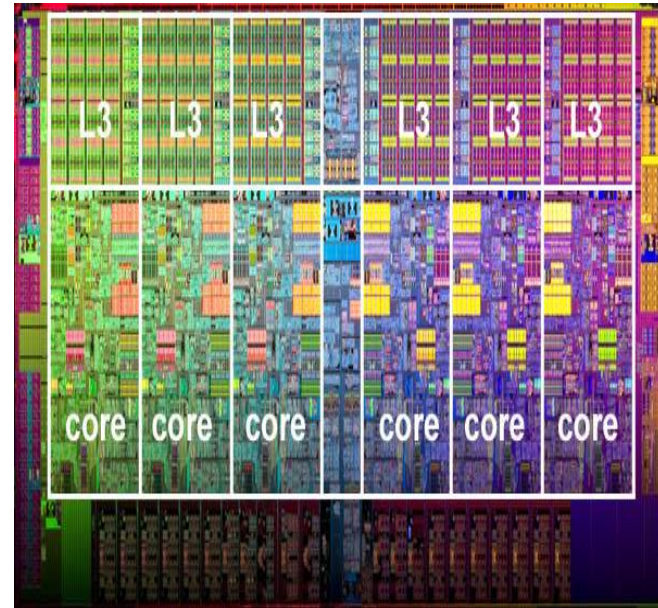
- Many problems are so large and/or complex that it is impractical or impossible to solve them on a single computer, especially given limited computer memory.
- Example: Web search engines/databases processing millions of transactions every second

PROVIDE CONCURRENCY (Main Reasons)

- A single compute resource can only do one thing at a time. Multiple compute resources can do many things simultaneously.
- Example: Collaborative Networks provide a global venue where people from around the world can meet and conduct work "virtually".

MAKE BETTER USE OF UNDERLYING PARALLEL HARDWARE (Main Reasons)

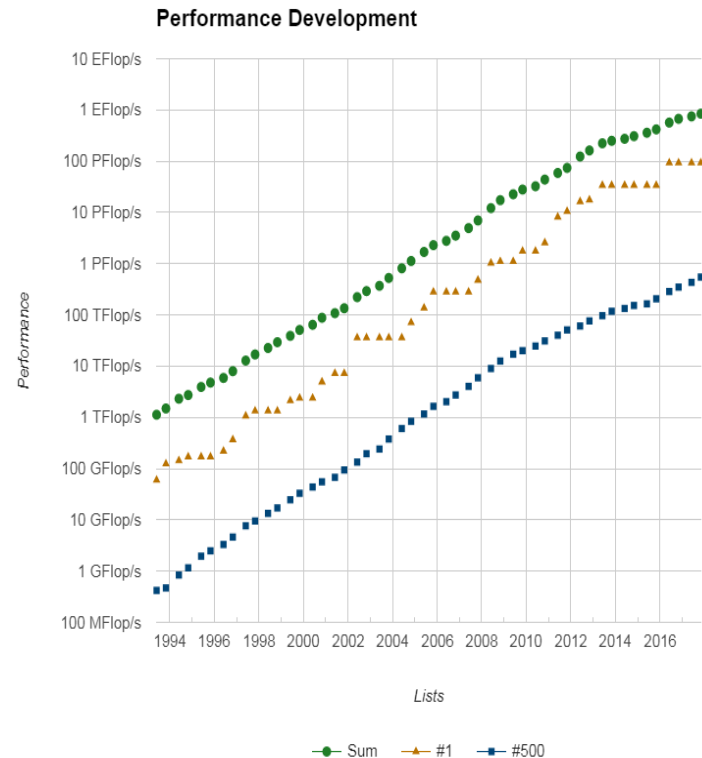
- Modern computers, even laptops, are parallel in architecture with multiple processors/cores.
- Parallel software is specifically intended for parallel hardware with multiple cores, threads, etc.
- In most cases, serial programs run on modern computers "waste" potential computing power.



Intel Xeon processor with 6 cores and 6 L3 cache units

The Future (Main Reasons)

- During the past 20+ years, the trends indicated by ever faster networks, distributed systems, and multi-processor computer architectures (even at the desktop level) clearly show that ***parallelism is the future of computing.***
- In this same time period, there has been a greater than **500,000x** increase in supercomputer performance, with no end currently in sight.
- ***The race is already on for Exascale Computing!***
- Exaflop = 10^{18} calculations per second





That's all for today!!